ENAE464 - 0202

Lab01: Pressure Measurements of Inviscid Flow over Cylinder

Due on February 13$^{\text{th}}$, 2026 at 11:59 AM

Dr. Silbaugh, 02:00 PM

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Experiment Performed: February 6$^{\text{th}}$, 2026
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In general, the letter should be addressed to the person or organization that requested the information contained in your report. For the purpose of this course, however, your letter can be addressed to the laboratory professor whose experiment you just have completed.

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3 Introduction

For the inviscid-flow model used in this lab, we assume a potential-flow field that is:

- **Incompressible (continuity):** $\nabla \cdot \vec{\mathbf{V}} = 0$.
- **Irrotational:** $\nabla \times \vec{\mathbf{V}} = 0$.

If $\nabla \times \vec{\mathbf{V}} = 0$, then a *velocity potential* ϕ exists such that $\vec{\mathbf{V}} = \nabla\phi$. Substituting into continuity gives Laplace's equation:

$$\nabla^2\phi = 0 \qquad \left(\text{i.e., } \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0\right).$$

In order to model flow over a 2D cylinder, we use an approach presented in Anderson's Fundamentals of Aerodynamics, which uses a superposition of a uniform flow and a doublet flow, as presented in the figure below:

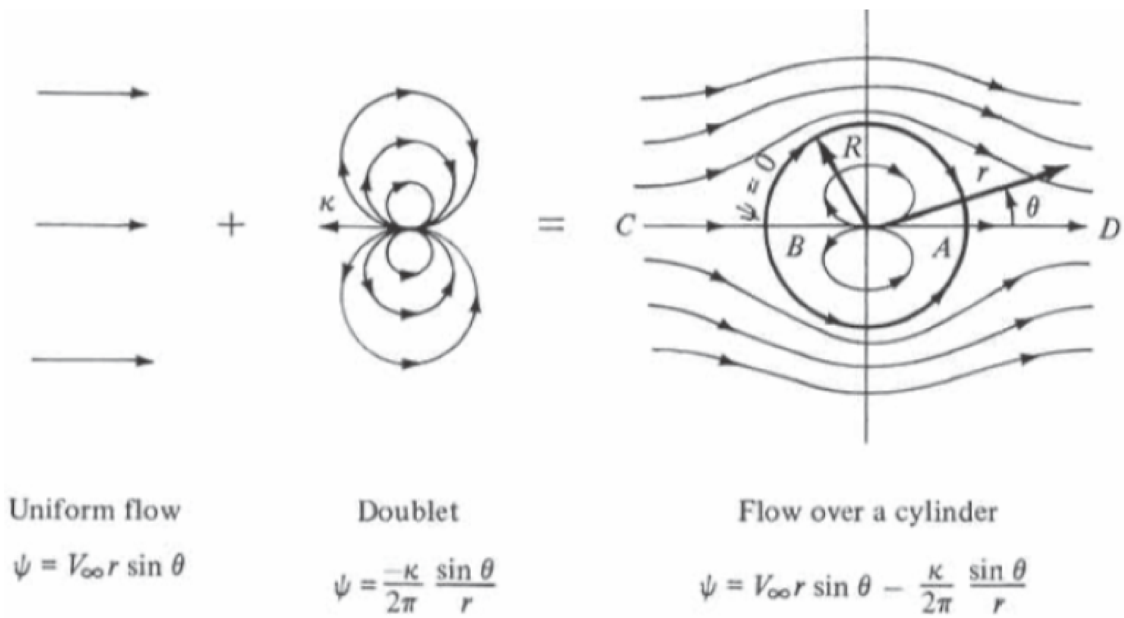

Figure 3.26 Superposition of a uniform flow and a doublet; nonlifting flow over a circular cylinder.

Figure 1: Superposition of uniform flow and a doublet to model nonlifting potential flow over a circular cylinder.

In polar coordinates (r, θ), the stream function and velocity potential for both the uniform flow and the doublet are given by:

$$\phi_U = V_\infty r\cos(\theta) \qquad \text{(velocity potential: uniform flow)} \tag{1}$$
$$\psi_U = V_\infty r\sin(\theta) \qquad \text{(stream function: uniform flow)} \tag{2}$$

$$\phi_D = \frac{\kappa}{2\pi}\frac{\cos(\theta)}{r} \qquad \text{(velocity potential: doublet)} \tag{3}$$

$$\psi_D = -\frac{\kappa}{2\pi}\frac{\sin(\theta)}{r} \qquad \text{(stream function: doublet)} \tag{4}$$

where V_∞ is the freestream speed and κ is the doublet strength.
Superposing the uniform flow and doublet stream functions gives

$$\begin{aligned}\psi &= \psi_U + \psi_D \\ &= V_\infty r\sin(\theta) - \frac{\kappa}{2\pi}\frac{\sin(\theta)}{r} \\ &= V_\infty r\sin(\theta)\left(1 - \frac{R^2}{r^2}\right),\end{aligned} \tag{5}$$

where the cylinder radius is defined by

$$R^2 \equiv \frac{\kappa}{2\pi V_\infty}.$$

The corresponding velocity field in polar coordinates follows from the stream function relations $V_r = \frac{1}{r}\frac{\partial\psi}{\partial\theta}$ and $V_\theta = -\frac{\partial\psi}{\partial r}$:

$$V_r = \frac{1}{r}\frac{\partial\psi}{\partial\theta} = \frac{1}{r}\left(V_\infty r\cos(\theta)\right)\left(1 - \frac{R^2}{r^2}\right) = \left(1 - \frac{R^2}{r^2}\right)V_\infty\cos(\theta), \tag{6}$$

$$V_\theta = -\frac{\partial\psi}{\partial r} = -\left(1 + \frac{R^2}{r^2}\right)V_\infty\sin(\theta). \tag{7}$$

At the surface of the cylinder, $r = R$, the velocity field is:

$$V_r = \left(1 - \frac{R^2}{R^2}\right)V_\infty\cos(\theta) = 0, \tag{8}$$

$$V_\theta = -\left(1 + \frac{R^2}{R^2}\right)V_\infty\sin(\theta) = -2V_\infty\sin(\theta). \tag{9}$$

This aligns with the assumption that the velocity normal to the stationary surface must be zero, and due to polar coordinates, the velocity normal to the surface is given by V_r. Since V_θ is perpendicular to the surface, due to construction of the coordinate system, it's equivalent to a tangential velocity, and since normal velocity is zero, $V = V_\theta = V_\infty\sin(\theta)$.
The pressure coefficient is given by:

$$C_p = 1 - \frac{V^2}{V_\infty^2}$$

$$V = -2V_\infty\sin(\theta) \tag{10}$$

$$\frac{V^2}{V_\infty^2} = 4\sin^2\theta \tag{11}$$

$$\boxed{C_p = 1 - \frac{V^2}{V_\infty^2} = 1 - 4\sin^2\theta} \tag{12}$$

4 Experimental Apparatus and Procedures

4.1 Experimental Apparatus

4.2 Operating Technique

5 Results

5.1 Operating Conditions

Table 1: Ambient Room Conditions during Experiment

Quantity	Value	Uncertainty
Room Temperature, T	25 °C	±0.5 °C
Room Pressure, P	1009.0 hPa	–

The density of air (ρ) can be calculated using the Ideal Gas Law:

$$\rho = \frac{P}{RT}$$

where:

- P is the absolute pressure (in SI units: Pa)
- T is the absolute temperature (in Kelvin)
- R is the specific gas constant for dry air, $R = 287.05\ \mathrm{J/(kg \cdot K)}$

Given measurements:

$$\begin{aligned}
P &= 1009.0\ \mathrm{hPa} = 100900\ \mathrm{Pa} \\
T &= 25\ ^\circ\mathrm{C} = 25 + 273.15 = 298.15\ \mathrm{K} \\
R &= 287.05\ \mathrm{J/(kg \cdot K)}
\end{aligned}$$

$$\rho = \frac{100900\ \mathrm{Pa}}{287.05\ \mathrm{J/(kg \cdot K)} \times 298.15\ \mathrm{K}}$$

Calculate:

$$\begin{aligned}
\rho &= \frac{100900}{287.05 \times 298.15} \\
&= \frac{100900}{85542.5} \\
&\approx 1.18\ \mathrm{kg/m^3}
\end{aligned}$$

Thus, **the ambient air density during the experiment is**:

$$\boxed{\rho = 1.18\ \mathrm{kg/m^3}}$$

5.2 Measurements

5.3 Quantities of Interest

The theoretical value of drag coefficient is 0 due to the fact that with the assumption of inviscid flow $c_f = 0$ and symmetry of the cylinder shape, and the symmetry of the flow around it which results in equal pressure on opposing sides canceling each other out.
Performing numerical analysis using the equations from the lecture:

$$C_D = \frac{D}{\rho_\infty}$$
$$D = R \int_0^{2\pi} P \cos(\theta)\, d\theta$$

Due to cylinder symmetry:
$$D = R \int_0^{2\pi} P \cos(\theta)\, d\theta = 2R \int_0^{\pi} P \cos(\theta)\, d\theta$$

To numerically calculate the integral, we used the trapezoidal approximation, and the results of C_D vs. θ are shown in the figure below:

6 Analysis and Discussion

As can be seen in the figure, the actual coefficient of drag is not zero due to the viscosity of the air and flow detachment.

6.1 Data Conversion

6.2 Interpretation

7 Summary and Conclusions

8 References

References

[1] J. D. Anderson and C. P. Cadou, *Fundamentals of Aerodynamics*, 7th ed. McGraw-Hill Education, March 17, 2023, pp. 209–324, ISBN: 9781266076442.

9 Appendices

9.1 Data

`./data/pressure_vs_alpha.csv`

	α [°]	P_∞ [cm]	P_0 [cm]	P [cm]
01	0	15.4	23.0	23.0
02	5	15.4	23.1	22.7
03	10	15.4	23.1	22.3
04	15	15.4	23.2	21.3
05	20	15.3	23.1	19.9
06	25	15.4	23.1	18.6
07	30	15.4	23.2	16.9
08	35	15.4	23.1	15.0
09	40	15.4	23.1	13.1
10	45	15.4	23.1	11.2
11	50	15.3	23.1	9.4
12	55	15.3	23.1	8.4
13	60	15.4	23.1	7.4
14	65	15.4	23.1	7.0
15	70	15.4	23.2	6.9
16	75	15.4	23.2	7.6
17	80	15.4	23.2	7.8
18	85	15.4	23.1	8.0
19	90	15.4	23.1	8.1
20	95	15.4	23.1	8.3
21	100	15.4	23.2	8.3
22	105	15.4	23.1	8.2
23	110	15.4	23.1	8.1
24	115	15.4	23.1	8.1
25	120	15.4	23.1	8.1
26	125	15.4	23.1	7.9
27	130	15.4	23.1	7.8
28	135	15.4	23.1	7.8
29	140	15.4	23.1	7.7
30	145	15.4	23.1	7.6
31	150	15.4	23.1	7.7
32	155	15.4	23.1	7.8
33	160	15.4	23.0	8.0
34	165	15.4	23.1	8.0
35	170	15.4	23.1	8.2
36	175	15.4	23.1	8.2
37	180	15.4	23.1	8.2

Table 2: Raw Measurements of Pressure Taps vs. Angle of Attack

9.2 Code